X's feed algorithm is value misaligned for both Democrats and Republicans, but more so for Democrats

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Abstract

Social media feed algorithms rank content that is purported to be preferred by users, but the engagement signals that drive these algorithms are (at best) an indirect proxy of users' preferences and underlying values. Here, we investigate the consequences of ranking algorithms on expressions of basic human values on X (Twitter) by annotating the values expressed in participants' social media feeds ($N = 715$ Americans), analyzing the relationship between the posts' value expressions and the posts' amplification in the ranked "For You" Page (FYP) feed, and then comparing the amplified values to users' own values. We observe that users' followed accounts (inventories) reflect their own values—but that there is an overall negative correlation (misalignment) between users' own values and what the algorithm amplifies. We also detect partisan differences: the values the algorithm amplifies are negatively correlated with both Democrats' values and Republicans' values, but more negatively correlated for Democrats. However, despite this overall negative association, both Democrats and Republicans engagement behaviors are broadly correlated with their own values, resulting in a "value gap" between the purported objective of the algorithm and the empirical behavior of the algorithm maximizing that objective.

Keywords: Social media | algorithmic personalization | media amplification | value alignment

Introduction

Algorithms that rank social media feeds take into consideration a set of posts including those from accounts that an individual follows—their in-network inventory—making a personalized selection from this inventory. These feed algorithms are typically optimized for engagement, promoting and demoting content to maximize an objective that weighs the user's predicted clicks, likes, and replies [2, 11, 3]. Concerns over the spread of misinformation, the promotion of toxic content, and the formation of polarized 'filter bubbles' have raised the question about the perverse effects of this optimization for engagement (see, e.g., [7]). Much of the effort to measure what content social media algorithms promote or demote ("algorithmic amplification") has focused on political valence [9, 20] and emotionality [10, 1]. However, the field of value alignment has also demonstrated that a broader concern lurks: that the values encoded in these algorithms may diverge from what people actually value [6]. Here, we address this by focusing on the association between a user's values and those that the algorithm is promoting, i.e., "the value alignment" of social media algorithms with their users. This paper provides systematic quantitative insights into the behavior of engagement-based ranking algorithms through the lens of *basic human values*. We draw on Schwartz's refined theory of Basic Human Values [15], which articulates a set of 19 human values (see Fig. 1) that nest into higher-level categories and has been validated across cultures [15]. We leverage a validated method for annotating the expressions of basic human values in X/Twitter posts [4] (SI Sec. 2) to measure the values associated with algorithmic promotion from a user's inventory into their algorithmic "For You" feed (FYP), enabling the study of what we term *value amplification*.

We recruited a nationally representative sample of $N = 715$ Americans who are active users of X in September and October 2024, quota matched on ethnicity, gender and partisanship to install a browser extension to collect posts from both their FYP and the Following feed which forms the in-network inventory (see SI Sec. 1). To measure value amplification, we regress amplification from the values expressed in a post over all in-network inventory posts (see Methods and SI Sec. 3).

Results

First, we identify which values are amplified by the FYP algorithm. At the highest level, we observe more amplification of individualist values ($\beta = 0.023$) than collectivist values ($\beta = 0.002, p < 0.001$). However, when considering the four middle level clusters that comprise individualism and collectivism (*Self-transcendence*, *Openness to change*, *Conservation*, *Self-enhancement*, see Figure 1 top), we see heterogeneity: on one hand the individualistic cluster of *Openness to change* is amplified ($\beta = 0.034, p < 0.001$) while the individualistic cluster of *Self-enhancement* is demoted (de-amplified, $\beta = -0.036, p < 0.001$). And on the other hand the collectivist cluster of *Conservation* is amplified ($\beta = 0.133, p < 0.001$) while the collectivist cluster of *Self-transcendence* is demoted ($\beta = -0.0541, p < 0.001$).

When considering the low-level values (see Figure 1 bottom), we observe the highest rates of amplification for conservation values such as rule conformity (β=0.0228), societal security (β=0.02143) and tradition (β=0.0172). Meanwhile, we also observe de-amplification (i.e. demotion) of self-transcendent values, such as caring (β=-0.012), dependability (β=-0.00923), as well as universal concern (β=-0.0056) and preservation

of nature (β=-0.0029).

But whose values are amplified? In Fig. 2A, we compare individuals' own values to the values expressed in their social media inventory and the values amplified by their feed ranking algorithms by computing the mean Spearman rank correlation between each individual's personal values and the frequency of values in user's in-network inventory, the amplification coefficient for each value, and the value-level coefficients estimated for engagement (see SI Sec. 4). We observe that individuals' values are broadly aligned with the values represented in their inventories ($\bar{\rho} = 0.302$, 95% CI=$[0.2837, 0.3197]$, meaning that people follow accounts that generally produce content aligned with their own values. This result does not meaningfully differ across political identity (t-test, $t = 0.0011$ $p = 0.999$).

However, we observe an overall negative correlation between participants' values and the likelihood of algorithmic amplification in a post ($\bar{\rho} = -0.150$, 95% CI=$[-0.169, -0.132]$), which suggests that the algorithm is actually amplifying content that is *misaligned* with users' values. We observe heterogeneity by political identity: Democrats' amplification ($\bar{\rho} = -0.22917$) is more misaligned with their values than Republicans' ($\bar{\rho} = -0.041$, $p < 0.001$).

We find that the values that people *perceive* are amplified by the algorithm follow the same partisan spread pattern as actual amplification, with Democrats perceiving the algorithm as less aligned with their values than Republicans ($\bar{\rho}_{\text{Rep}} = 0.162, \bar{\rho}_{\text{Dem}} = 0.031$, $p_{\text{diff}} < 0.001$) but the extent of the actual misalignment is significantly greater than people think.

As a robustness check, we conduct a pre-registered replication study that largely found the same pattern of results as the main study (see SI Section 6). While the replication also found significant partisan differences in alignment between users and amplification ($p < 0.001$), we found a small but positive alignment for Republicans ($\rho = 0.0372$) and a negative alignment for Democrats ($\rho = -0.199$), pointing to the stronger claim that X's algorithm could be aligned for Republicans but misaligned for Democrats (see SI Section 6).

Explaining value-misaligned amplification

Despite the fact that users' inventories are aligned with their own values, the algorithm actually amplifies content that is *misaligned* with their values. What could explain this pattern of results? Since the algorithm explicitly learns from engagement behavior, we leverage two datasets of engagement behavior to provide two possible mechanisms behind the "value gap" between the purported engagement objective of the algorithm and the empirical behavior of the algorithm maximizing that objective (see Materials and Methods): 1) actual engagement behavior for our participants during the 24 hours after they installed the Chrome extension (personal), and 2) the overall engagement metrics of posts in participants' feed (global).

For individual favoriting and reposting, we find high levels of value-alignment for both Democrats ($\bar{\rho}_{\text{Fav}} = 0.254, \bar{\rho}_{\text{Repost}} = 0.408$) and Republicans ($\bar{\rho}_{\text{Fav}} = 0.222, \bar{\rho}_{\text{Repost}} = 0.284$), Fig. 2B). However, we also find suggestive evidence that replies are less value-aligned than these other forms of engagement, and are in fact even value-misaligned on average for Democrats ($\bar{\rho}_{\text{Dem}} = -0.0702, \bar{\rho}_{\text{Rep}} = 0.056$). Despite the fact that replying is a sparse event (only 6.8% of engagement events), it is known that X weights predicted replies more heavily than predicted favorites (up to 27× more [18]). Thus, it is possible

that this value gap for favorites, and partisan difference in replies, may steer the algorithm to differentially amplify value-misaligned content for Democrats. This could create a feedback loop, where value-misaligned content is shown to these users, who reply to it, thus amplifying more content for future replies.

We find that the values associated with *global* engagement on the platform (number of favorites, retweets, replies) are not only less value aligned than the values of posts individuals engage with *personally* (for both retweets and favorites), but also that they are systematically more aligned with Republican users than Democrat users (Fig. 2B, star). This suggests that the algorithm may be learning from global engagement on the platform, which is principally constituted by posts espousing Republican values and therefore preferentially aligns with Republican users.

Discussion

We observe a core tension with how engagement-maximizing algorithms operate on social media: on average, X's feed algorithm amplifies engaging content, but ultimately, this content is misaligned with many users' own values, particularly Democrats. This aligns with recent work that shows that switching from a chronological to an algorithmic feed shifts political opinion towards more conservative positions [7]. We also reveal a possible mechanisms in differential patterns in how Democrats and Republicans consume information, and engage with that information, in turn leading to divergent experiences and asymmetries in online discourse. This approach offers new ways to evaluate and understand social media ranking algorithms without internal access to the platforms, connecting in-network amplification with constructs of societal relevance (see SI Sec.7 for limitations and future work).

This novel measurement framework raises the question: whose values should these algorithms align with? If social media is to be an open-minded "town square," then perhaps values should be amplified in proportion to their frequency ($\beta = 0$). But if instead social media is to reflect its users values, then it may amplify a distinct set of values. These two visions of social media algorithms require different paradigms for content moderation, regulation, and user control.

Beyond its empirical findings, this work also highlights the possibility of "value-based ranking," where feed algorithms could be steered towards users' values. On one hand, this may be a way to connect revealed preferences and stated preferences [10] and identify *bridging values* that span ideology [12]. On the other hand, it may reinforce value-based echo chambers. Ultimately, future work is required to assess the feasibility of value-based ranking and its effects on users.

Materials and Methods

This work has been approved by Stanford IRB protocol #IRB-68917.

Measuring basic human values

We annotate the values in posts using a fine-tuned LLM on majority consensus on 600 posts selected for high levels of agreement from a separate sample of $1,079$ nationally representative participants [4] (see SI Sec. 2). Annotations are then personalized at the

level of each participant to account for population-level disagreements in annotation. For amplification, our primary pre-registered regression models use as independent variables the first four principal components of a PCA for the 19 low-level values (these four principal components explain 90.37% of variance across these values) and random effects for users (see SI Sec. 3).

Individual engagement

We collect engagement events (likes, reposts, and replies) from our participants, from browser plugin observation for 24 hours after feed collection. We observe 31,536 view events from N=1,023 participants and 961 unique engagement events from 149/1,023 participants that have corresponding value labels. We predict engagement as a function of the Schwartz values expressed from viewed posts (see SI Sec. 5A).

Global engagement

We collect the overall engagement metrics (number of likes, reposts, and replies) from posts in our participants in-network inventory. We normalize these counts by log transforming them and predict engagement as a function of the Schwartz values expressed from inventory posts (see SI Sec. 5B).

Author Contributions

ZE, FJ, and TP collected and curated the data with help from AP, IG, SS, DZ. ZE, FJ, TP, JU and MB analyzed the data. All authors wrote and reviewed the paper.

Competing Interests

The authors declare no competing interests.

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SI Section 1: Inventory Collection

We recruited 1000 participants via Prolific, to use our Chrome Extension in September and October 2024, across 3 waves of data collection (Sep 23-29, Oct 18 and Oct 29-Oct 31). Of those 1000, 715 made it to the post-data collection survey and we identified a match between their survey data and extension data. Participants were all from the United States and were required to be 18 years old or older. We recruited a nationally representative sample of social media users quota matched on ethnicity, gender, partisanship and age. However, we were unable to meet the nationally representative quota for participants in the age range 55-100 due to the fact that we required participants to have an active social media (X) account to participate. Nonetheless, 7.5% of our sample was over the age of 55. Within this sample, the median age was 43 (min=19, max=80),

58.36% White/Caucasian, 51.2% female and 30.6 %Democrat, 28.8% Republican and 40.6% Independent.

Participants were asked to install a Chrome Extension that loaded their X homepages and then used it to download posts from both their For You (FYP) and their Following feeds simultaneously. The Chrome Extension aims to collect 75 batches of tweets (groups of tweets that load together) from the FYP in addition to as many Following Batches as necessary until it encounters a tweet posted more than 7 days prior to the time of retrieval or as many as possible within 30 minutes, whichever happens sooner.

The rationale for the number and criteria for feed collection is as follows: Heavy Ranker—the recommender system used to rank the tweets in FYP—is fed as input approximately 1500 tweets that are selected from a pool of recent in-network and out-of-network tweets [17][1]. Then, at the last stage of the pipeline, promotions including ads and recommendations about who to follow are additionally mixed into the ordered set. With the conservative assumption that within each FYP batch there exist 20 non-promotional tweets, fetching 75 batches ensures that we have collected all the tweets output from one run of the Heavy Ranker. Our focus is on investigating which in-network tweets are amplified in the FYP. Because the Heavy Ranker only processes "recent" tweets, we also limit the Following feed collection to recent tweets. While the exact definition of "recent" is unclear, our pilot tests suggest that tweets older than 7 days do not appear in the FYP. Thus, we constrain the Following feed collection to tweets posted within 7 days prior to retrieval. We filter to participants for whom we identify at least 30 tweets that are algorithmically amplified from their in-network inventory to their For You page.

From the 715 participants, we analyzed 2,952,899 tweets, with an average of 4,130 posts from the Following page per user, of which on average 524 were amplified into the For You page. To capture context, the text of a given tweet (tweet_1) made in reply to or quoting another tweet (tweet_2) is represented as "tweet_1 REPLY TO: tweet_1" and "tweet_1 QUOTED: tweet_1" respectively.

Given the dataset of 2,952,899 following posts, of which 374,887 appear in FYP, from 715 participants, we construct an inventory dataset by constructing a binary indicator of whether or not each post in a given user's Following page appeared in the algorithmically ranked FYP page (`is_amplified`). After this restriction, we observe 4130 posts per person and an average amplification rate of 0.3618.

Given that we collect longer time frames of Following posts than For You posts, we limit our analysis only to a timeframe where we would expect amplification to occur. In particular, for a given user, we limit to all Following posts that appear in 6-hour time blocks where we observed nonzero amplification i.e. at least one post that appears in their For You feed, resulting in a dataset of 2,952,899 inventory posts across participants.

SI Section 2: LLM Model for Value Annotation

Once the inventories have been collected of our participants, we must then annotate the posts in these inventories for values. To do so, we use the method developed and explained in [5]. For convenience, we elaborate on the core components of this approach below.

[1] At the time of our study, approximately half of content in a X/Twitter user's "For You" feed was in-network (inventory) content, and the other half out-of-network content [19].

Sourcing a Dataset of Social Media Posts

First, we recruited a sample of social media users (N=281) quota-matched to be a representative sample of the US population on age, ethnicity, gender and partisanship using Prolific in May 2024. Participants were all from the United States and were required to be 18 years old or older. We were unable to meet the quota for participants in the age range 55-100 (92/109) so we upsample posts from participants in this age range by 18.4% when constructing our silverset of social media posts. Within this sample, the median age was 43 (min=19, max=80), 58.36% White/Caucasian, 51.2% female and 30.6% Democrat, 28.8% Republican and 40.6% Independent. Participants were paid $\sim$ \$15.42 an hour and took approximately 7 minutes to complete the study.

After they consented to the research, participants were asked to install a Chrome Extension that downloaded posts from their X (Twitter) feeds, both their algorithmically curated For You page (FYP) and their in-network Following feeds simultaneously. We collect and annotate posts from both the Following feed and FYP to account for any systematic differences that may exist between them.

When a browser opens the X page, tweets are fed to the browser client in "batches": a single API request returns a set of tweets at a time. The Chrome Extension aims to download 16 batches of tweets: 7 from the FYP, and 9 from the Following page. Each batch for FYP consists of approximately 30 tweets, whereas the Following feed returns a variable number of tweets up to 100.

From the 281 participants, we collected 265,442 posts, with an average of 207 posts from the FYP and 776 from the Following page per user. Among these 265,442 posts, there are 151,740 posts with unique, non-empty text in English (using Google's Chromium language detection tool `cld2`). We also screen out private posts (n=6,025; 3.97%), deleted posts (n=2,004; 1.32%) and/or reply tweets that are replying to posts not displayed in the user's feed (n=7,581; 4.99%) to focus only on the 142,652 public posts we can render with proper context.

We next prepare these posts for human annotation, since some posts are either inappropriate (NSFW) to show to annotators or incomprehensible to most annotators. We filter the posts for comprehensibility (grounded in the constructs of readability, coherence, spam behavior, and context required for understanding) and NSFW (to comply with our institution's IRB for data annotation) using an automated GPT-based pipeline, with the prompt that rates all 19 values jointly each on a 0-6 scale based on a short description with `GPT-4o`, temperature=1.0 and seed=0. We filter out any posts that score less than 3 on a comprehensibility scale, resulting in a dataset of 89,383 (62.6%) posts. We then filter out 1,937 posts with NSFW content, resulting in a final dataset of 87,446 posts.

The occurrence rate of different values in social media posts varies, as do the numbers of posts per person across both Following and FYP feeds. Therefore, uniform sampling procedure may yield a sample that underrepresents rarer values and overrepresents certain users for annotation. To construct a final set of posts for human annotation, we stratify on user and post source as well as on a preliminary LLM-based measure of value expression tuned for recall over precision to prioritize catching as many potential value-expressing posts as possible. Since stratification requires an initial estimate of which value a post may express, we first built an automated classifier tuned for recall rather than precision—that is, we prioritized sampling as many potential value-expressing posts as possible, even if this meant including some false positives.

In particular, we perform a preliminary rough classification of these posts for all 19 of

the Basic Human Values [15] on a 7-point Likert scale [0-6] using another automated GPT-based pipeline with `GPT-4o`, temperature=1.0 and seed=0. This allows us to upsample posts that may reflect rarer values. For this upsampling, we say a given post reflects a given value if the GPT annotation pipeline scored it a 4 or above. We then sample a final set of 5,227 posts for human annotation by stratifying across users, post source (FYP vs following), and value expression. In particular, for each user and each value, we sample a tweet from that user's feed that reflects that value. We sample FYP and Following posts with equal probability, but for a given user, if only one type of post exists, we return that tweet. Because we did not meet our quota for participants over the age of 55 (by 17 participants), we randomly sample (without replacement) 17 of the 76 participants over the age of 55 who completed the survey and repeat this process for each of them to ensure representation of posts from users over the age of 55. Of the 5,227, 3,182 came from users' Following feeds (60.8%) and 2,045 (39.1%) came from users' FYP.

This process yielded a dataset of 5,227 posts that we can use to study human value annotation and to predict the values expressed in those posts. In this section, we describe our ground truth data annotation method.

Annotating the Dataset of Social Media Posts

In total, 1,276 participants began the survey. We filtered out N=153 participants who failed two simple attention checks (see Appendix) or who did not finish the study. N=1,123 completed the study. Our final dataset included ratings from N=1,079 participants who have a full set of ratings for all 30 posts assigned to them and within this sample, the median age was 45 (min=18, max=85), 61.2% White/Caucasian, 50.8% female and 29.6% of participants lean Democrat vs 28.63% lean Republican. Participants were paid $15.00 an hour and took approximately 60 minutes to complete the study.

Each participant provided value expressions ratings for 30 posts. The posts for each participant were randomly sampled from the dataset of 5,227 posts sourced as described above (each post was rated by a median of 6 people). Given this sampling procedure, we retained ratings of 5,211 posts (since posts are sampled with replacement for each participant).

We used a recursive labeling where raters could traverse the value tree to elicit value ratings for all 19 values in an efficient manner. The branching method leverages the tree structure of the Schwartz Theory of Basic Values , starting with the high-level values, and traversing down only those branches whose parent values they had marked as expressed in the post.

Following this primary task, participants completed the Value Calibration Questionnaire (VCQ) of 25 additional ratings on a shared set of questions. Finally, participants answered a number of demographic and political questions.

To familiarize them with the task and to determine eligibility for proceeding to the main labeling task, participants first completed a short training for four synthetic posts followed by a gating task. The training introduced annotators to the meaning of the Schwartz values, while the gating task assessed their understanding. These training posts were created to unambiguously reflect particular values, and participants were not allowed to proceed until they checked the right answer. If they answered incorrectly, a message appeared giving them the correct answer and asking them to try again.

After training, they proceeded to the gating phase, where they rated four posts that, based on a pre-study, either unambiguously contained or did not contain the values of

self-direction action or face. Participants who got one or fewer of the four correct were not allowed to continue. We took care in selecting the questions for both pre-tasks and in setting the eligibility threshold for the gating task to ensure a shared baseline understanding.

After completing basic attention screeners and training, participants who successfully passed the gating phase proceeded to rate 30 posts drawn at random from the larger set of 5,227 posts for value expression using the branching method discussed above. When presenting a tweet to our annotators for value annotation, to convey context, the text of a given tweet (tweet_1) made in reply to or quoting another tweet (tweet_2) was represented as "tweet_1 REPLY TO: tweet_1" and "tweet_1 QUOTED: tweet_1" respectively. The annotation dataset we create is available for researchers on a case by case basis to preserve the privacy of the participants. It is stored securely in an de-identified manner.

Training and Evaluating the Model

Due to the subjective nature of values [5], we use a personalized model which predicts *the extent to which an individual perceives* a value in a social media post. This model uses post-level and individual-level features to predict value expression. The post-level features are the consensus predictions from a gpt-4o model fine-tuned on consensus annotations. The individual-level features are that individual's responses to the 25 VCQ questions. For each post–user pair, these models predict that user's 0–6 rating for a given value V. We train 19 random forest models (one per value) trained on the ratings of 3000 posts stratified by number of ratings (see [5] for details). This model has been shown to outperform both off-the-shelf LLM methods and fine-tuned methods [5]. We use these random forest models to annotate the inventory posts for values.

SI Section 3: Amplification Analysis

The Following feed is a reverse-chronological, complete inventory of all in-network content that was available to their For You algorithm. For each participant we collect 4,130 inventory posts on average and identify the subset of which that appear in that user's FYP. Despite the fact that we collect in-network inventory posts from a week or more prior to data collection, we observe recency effects for algorithmic amplification, whereby only recent posts are eligible for amplification. Therefore, we subset to all in-network inventory posts that appear in 6-hour time blocks where we observed nonzero amplification i.e. at least one post that appears in their For You feed, resulting in a dataset of 2,952,899 inventory posts across participants. To account for recency effects of the algorithm (where only recent inventory posts are amplified), we restrict to 6-hour time blocks in which at least one inventory post was amplified into the For You feed, resulting in a dataset of 2,952,899 inventory posts across participants.

Our pre-registration can be found at `https://aspredicted.org/cg9n-rpqc.pdf`. Our primary analysis involves a principal component analysis of the 19 factors. We observe that 4 factors explain 90.37% of variation. We observe that PC1 describes the self/other highest order level of the Schwartz framework with the highest positive loadings on the values of caring (0.361), universal concern (0.354), dependability (0.331), humility (0.274) and tolerance (0.239), and the largest negative loadings on the values of hedonism (-0.3765), self-directed action (-0.264), self-directed thought (-0.2618), stimulation

(-0.2526), and achievement (-0.149) in the rotation matrix. PC2 describes the higher level value of conservation in contrast to the self/other highest order level, with the highest positive loadings similar to PCA1 on the values of caring (0.3539), dependability (0.305), nature (0.2644), hedonism (0.264) and universal concern (0.25924), but having the largest negative loadings on the conservation values of rule conformity (-0.3691), societal security (-0.3339), tradition (-0.2444), interpersonal conformity (-0.200), and dominance (0.008). PC3 also describes the higher level value of conservation, with the highest positive loadings on the values of rule conformity (0.375), societal security (0.366), tradition (0.337), interpersonal conformity (0.284) and self-directed actions (0.276), and the largest negative loadings on values of caring (-0.009), dependability (0.047), universal concern (0.112), nature (0.136), and personal security (0.152). PC4 distinguishes the higher level values of openness to change and self-enhancement, with the highest positive loadings on the values of achievement (0.548), face (0.459), resources (0.360), dominance (0.254) and personal security (0.0822), and the largest negative loadings on values of self-directed thoughts (-0.307), self-directed actions (-0.304), stimulation (-0.227), rule-conformity (-0.126), and societal security (0.118).

Our primary analysis involves regressing amplification of a post for a person on the principal components of that post for that person with a mixed-effects logistic regression model. The negative statistically significant coefficient on PC1 ($\beta = -0.0051, z = -5.333, p < 0.001$) thus indicates positive amplification of values related to the self, while the negative statistically significant coefficient on PC2 ($\beta = -0.0279, z = -14.964, p < 0.001$) indicates positive amplification of the higher order value of conservation. The positive statistically significant coefficient on PC3 ($\beta = 0.03080, z = 10.470, p < 0.001$) further indicates positive amplification of the higher order value of conservation. The negative statistically significant coefficient on PC4 ($\beta = -0.0147, z = -4.955, p < 0.001$) indicates positive amplification of the higher order value of openness to change.

To understand how these overall β from the principal component analysis map onto the low-level values, we can multiply the rotation matrix describing the loadings above by the β coefficients to reconstruct individual β's for low-level values. Using this approach, we identify a positive amplification of rule conformity ($\beta = 0.02286$), societal security ($\beta = 0.02144$), tradition ($\beta = 0.01728$), interpersonal conformity ($\beta = 0.01449$), self-directed thoughts ($\beta = 0.00833$), self-directed action ($\beta = 0.00833$), stimulation ($\beta = 0.00564$), personal security ($\beta = 0.00312$), dominance ($\beta = 0.00194$), hedonism ($\beta = 0.00160$). We observe a negative amplification of caring ($\beta = -0.01280$), dependability ($\beta = -0.00924$), universal concern ($\beta = -0.00561$), preservation of nature ($\beta = -0.00293$), achievement ($\beta = -0.00253$), tolerance ($\beta = -0.00210$), humility ($\beta = -0.00042$), face ($\beta = -0.00023$), resources ($\beta = -0.00010$).

An additional primary analysis involves regressing amplification of a post for a person on the highest level value clusters of that post (self/other for that person with a mixed-effects logistic regression model. The negative statistically significant coefficient on X ($\beta = -0.0051, z = -5.333, p < 0.001$) thus indicates positive amplification of values related to the self (e.g. the left side of the wheel in Figure 1).

An additional primary analysis involves regressing amplification of a post for a person on the higher level value clusters of that post (self transcendence, self-enhancement, conservation+ and openness to change) for that person with a mixed-effects logistic regression model. With this specification, we found a negative statistically significant coefficient on self-transcendence ($\beta = -0.0541, z = -12.638, p < 0.001$), a negative statistically significant coefficient on self-enhancement ($\beta = -0.0361, z = -5.196, p < 0.001$), a positive

statistically significant coefficient on conservation ($\beta = 0.1331, z = 18.970, p < 0.001$), and a positive statistically significant coefficient on openness to change ($\beta = 0.0343, z = 6.351, p < 0.001$).

A complementary approach to our primary analysis involves regressing amplification of a post for a person on all of the low level values of that post for that person with a mixed-effects logistic regression model. Using this approach, we identify the significant positive amplification of self-directed thoughts ($\beta = 0.1056, p < 0.001$), interpersonal conformity ($\beta = 0.046, p < 0.001$), face ($\beta = 0.04172, p < 0.001$), societal security ($\beta = 0.0309, p = 0.0134$), resources ($\beta = 0.0387, p = 0.0155$), significant negative amplification of dominance ($\beta = -0.0827, p < 0.001$), self-directed action ($\beta = -0.0510, p = 0.0015$), tolerance ($\beta = -0.0263, p = 0.0089$), hedonism ($\beta = -0.01871, p = 0.029888$), and no effect on achievement ($\beta = -0.00793, p = 0.4495$), stimulation ($\beta = -0.0112, p = 0.40044$), personal security ($\beta = -0.0082, p = 0.3954$), tradition ($\beta = 0.0062319, p = 0.469$), rule conformity ($\beta = 0.000378, p = 0.97241$), humility ($\beta = 0.0132, p = 0.189960$), dependability ($\beta = -0.0213434, p = 0.07902$), caring ($\beta = -0.01282, p = 0.2032$), universal concern ($\beta = 0.0089, p = 0.381$), nature ($\beta = 0.00335, p = 0.4577$)

These results are suggestive but ultimately the collinearity of the low-level model and the nature of PCA make direct low-level value analysis hard to interpret. In order to present maximally interpretable and precise measurements of low-level value amplification, we also present a modification of the higher level value analysis presented above that, for each of the 19, values, includes a low-level value marginal for each of the 19 values separately. In particular, for a given low-level value such as stimulation, we regress the binary indicator for post-level amplification or engagement against the extent to which that post expressed the high-level values of self-transcendence, openness to change, self-enhancement and conservation, and the marginal from subtracting stimulation from openness to change. This process is then repeated 19 times for each low-level value. Then to compute the overall coefficient and significance of the low level value, the coefficient on the marginal (stimulation-openness to change) is added to the coefficient on openness to change. This specification avoids problems of co-linearity while allowing us to identify the unique contribution of low-level values relative to each other.

Using this approach, we identify the significant positive amplification of interpersonal conformity ($\beta = 0.151, p < 0.001$), societal security ($\beta = 0.133, p < 0.001$), rule conformity ($\beta = 0.130, p < 0.001$), tradition ($\beta = 0.130, p < 0.001$), personal security ($\beta = 0.121, p < 0.001$), self-directed thoughts ($\beta = 0.118, p < 0.001$), face ($\beta = 0.072, p < 0.001$), self-directed action ($\beta = 0.0544, p < 0.001$), humility ($\beta = 0.0468, p < 0.001$), significant negative amplification of dominance ($\beta = -0.087, p < 0.001$), dependability ($\beta = -0.056, p < 0.001$), caring ($\beta = -0.051, p < 0.001$), tolerance ($\beta = -0.047, p < 0.001$), universal concern ($\beta = -0.0379, p < 0.001$), hedonism ($\beta = -0.0355, p < 0.001$), preservation of nature ($\beta = -0.0265, p < 0.001$), and achievement ($\beta = -0.0227, p < 0.001$), and no effect on resources ($\beta = -0.000, p = 0.608$) and stimulation ($\beta = -0.0166, p = 0.183$).

These results largely align with the amplification results presented in the main paper.

SI Section 4: Personal Values and Value Alignment

To measure individual values, we used the refined 57-item Portrait Values Questionnaire (PVQ) [14] to compute centered value scores for each participant for each of the 19 values [16]. Following the PVQ formulation, we take the average of the three sub-items for each of the 19 values and normalize within participant by subtracting each participants overall average value.

We measure value alignment using the average Spearman rank correlation between each individual's personal values and the frequency of values in the user's in-network inventory, the value-level regression coefficients estimated for amplification, and the value-level regression coefficients estimated for engagement.

We chose Spearman rank correlation for several reasons. For one, a correlation metric allows us to compare the *relative* expression across all 19 low-level Schwarz values, which follows the relational nature of the value system itself. Further, a correlation computation across values is inherently scale-invariant, allowing us to account for individual level differences in how the scale is used. This also allows us to compare individual values to several different value structures (algorithmic amplification vs inventory distribution vs values associated with engagement) all on the same scale. We chose Spearman rank correlation because the rank ordering captures the ordinal prioritization of values consistent with the underlying theory of basic human values. In addition, a correlation metric handles the ordinal nature of the value annotations, which exist on a 7-point Likert scale.

Results and Robustness Checks

Our primary analysis uses the amplification coefficients from the PCA model. With those values, we observe overall negative correlation between participants' values and the likelihood of algorithmic amplification in a post ($\bar{\rho} = -0.150$, 95% CI=$[-0.169, -0.132]$), which suggests that the algorithm is actually amplifying content that is *misaligned* with users' values. We observe heterogeneity by political identity: Democrats' amplification ($\bar{\rho} = -0.22917$) is more misaligned with their values than Republicans' ($\bar{\rho} = -0.041$, $p < 0.001$).

We also explore the extent to which the main results are robust to different approaches to measuring values. For both of the alterative approaches (marginal and collinear, below), we find qualitatively identical results as the primary analysis).

An alternative approach to measuring values is the marginal method (see Section 3). With those values, we observe overall negative correlation between participants' values and the likelihood of algorithmic amplification in a post ($\bar{\rho} = -0.0569$, 95% CI=$[-0.075, -0.038]$), which suggests that the algorithm is actually amplifying content that is *misaligned* with users' values. We similarly observe heterogeneity by political identity: Democrats' amplification ($\bar{\rho} = -0.1229$) is more misaligned with their values than Republicans' ($\bar{\rho} = 0.035$, $p < 0.001$).

A final alternative approach to measuring values is the collinear method (see Section 3). With those values, we observe overall negative correlation between participants' values and the likelihood of algorithmic amplification in a post ($\bar{\rho} = -0.06522$, 95% CI=$[-0.080, -0.0503]$), which suggests that the algorithm is actually amplifying content that is *misaligned* with users' values. We similarly observe heterogeneity by political identity: Democrats' amplification ($\bar{\rho} = -0.097$) is more misaligned with their values than Republicans' ($\bar{\rho} = -0.020838$, $p < 0.001$).

Variation by Political Posts

We used a political classifier to identify which posts in our dataset are political in nature. The classifier is a distilled language model from fine-tuning a RoBERTa model to identify political posts, including mentions of officials and activists, social issues, news, and current events, based on an annotated training set [13]. Within our amplification dataset, 37.29% are political in nature, and we repeat our amplification and alignment analyses for political and non-political content separately.

For political content, we observe overall negative correlation between participants' values and the likelihood of algorithmic amplification in a post ($\bar{\rho} = -0.056$, 95% CI=$[-0.0752, -0.0373]$), which suggests that the algorithm is actually amplifying content that is *misaligned* with users' values. We similarly observe heterogeneity by political identity: Democrats' amplification ($\bar{\rho} = -0.0989$) is more misaligned with their values than Republicans' ($\bar{\rho} = 0.0033$, $p < 0.001$).

For non-political content, we also observe overall negative correlation between participants' values and the likelihood of algorithmic amplification in a post ($\bar{\rho} = -0.088$, 95% CI=$[-0.106, -0.071]$), which suggests that the algorithm is actually amplifying content that is *misaligned* with users' values. We similarly observe heterogeneity by political identity: Democrats' amplification ($\bar{\rho} = -0.143$) is more misaligned with their values than Republicans' ($\bar{\rho} = -0.0125$, $p < 0.001$).

This suggests that our core results (a gap in value alignment for algorithmic amplification) between Republicans and Democrats holds for both political and non-political content and is not driven just by political content.

SI Section 5: Engagement Analyses

5A: Personal Engagement

We use a dataset of engagement events from our participants, collected through browser plugin observation for a period of 24 hours after their initial feed collection. We log engagement events for views, likes, reposts, and replies. In particular, in the 24 hours after feed collection , we observe 31,536 view events from N=1,023 participants and 961 unique engagement events from N=149/1,023 participants that have corresponding value labels. We conduct regression analysis mirroring those for amplification for each engagement action (reply, favorite, repost). In particular, we conduct a regression analysis over posts individuals viewed, predicting a binary indicator of engagement action as a function of the Schwartz values expressed in a post, with random effects for users. We similarly annotate the values in posts using a fine-tuned large language model trained on consensus annotations from a separate sample of 1,079 nationally representative participants [5] and then personalizing at the level of each participant using the personalization procedure outlined above in SI Section 2.

5B: Global Engagement

Within our amplification dataset, we also collect the overall engagement metrics (number of likes, reposts, and replies) from each post in our participants in-network inventory. We normalize these metrics by log transforming them and conduct a similar regression

analysis over inventory posts, predicting log engagement as a function of the Schwartz values expressed in a post, with random effects for users.

SI Section 6: Conceptual Replication of Alignment Analysis

To confirm the robustness of our main results, we conducted a conceptual replication with N=264 recruited from Prolific and quota matched on age, gender and ethnicity. This sample had a mean age of 43.53 years, 72.3% white, and 49.6% female and all identified that they use X/Twitter.

For each of these responses, we compute their personal values using the standard PVQ, and then compute their value alignment with the pre-registered metrics from Study 1. Our pre-registration for this replication can be found at `https://aspredicted.org/km4ra5.pdf`.

Overall, we find that algorithmic amplification is value misaligned with these participants ($\bar{\rho} = -0.1041$, 95% CI=$[-0.136, -0.072]$). We found that algorithmic amplification was value aligned for Republicans ($\rho = 0.0372$) and value mis-aligned for Democrats ($\rho = -0.199$) and the difference between the two was statistically significant ($p < 0.001$, pre-registered).

As specified in our pre-registration, we also look at alignment between participant's values and other stages of the social media funnel in the previously pre-registered data, such as base distributions of values in in-network inventory (Following feed), and posts the original participants engaged with 24 hours after the extension was installed (using the engagement model specification listed in the previous pre-registration).

For in-network inventory, we found that we find that the content of the Following feeds is value aligned with these participants ($\bar{\rho} = 0.2649$, 95% CI=$[0.237, 0.283]$). This was true for both Republicans ($\rho = 0.237$) and for Democrats ($\rho = 0.283$), with no statistically significant difference between the two ($p = 0.117$).

For individual engagement, we found similar patterns to the main analysis.

Individual favoriting was value aligned for both Republicans ($\rho = 0.24266$) and for Democrats ($\rho = 0.2090$), with no statistically significant different between the two ($p = 0.2645$).

For reposting, we found value alignment for both Republicans ($\rho = 0.2064$) and for Democrats ($\rho = 0.3695$), with a statistically significant difference between the two ($p < 0.001$). For replies, we found that replying were value aligned for both Republicans ($\rho = 0.090946$) but value misaligned for Democrats ($\rho = -0.0564$), and the difference between the two was statistically significant ($p < 0.001$, pre-registered).

Following our pre-registration, we also conduct the same analyses but using two different methods for analyzing values - a collinear methods and a marginal method (see pre-registration). The results are largely similar, exhibiting the same qualitative trends with a few minor differences.

For the marginal method, the main result of a partisan difference in alignment for algorithmic amplification holds, as we also find that algorithmic amplification was value aligned for Republicans ($\rho = 0.0816$) and value mis-aligned for Democrats ($\rho = -0.0894$) and a statistically significant difference ($p < 0.001$, pre-registered).

Because the value predictions are highly collinear by construction, the joint specification suffers from multicollinearity, inflating standard errors and leading to unstable coefficient estimates. This results in less precise estimates compared to the other two

specifications. For the collinear method, the main result of a partisan difference in alignment for algorithmic amplification is qualitatively similar but not significant ($p = 0.325$): we find that algorithmic amplification was misaligned for Republicans ($\rho = -0.039$) and value mis-aligned for Democrats ($\rho = -0.062$).

SI Section 7: Discussion, Limitations and Future Work

Feed algorithms are designed to recommend content to users that they are likely to engage with, an objective that requires some level of personalization of the content to the preferences of the user themselves. Yet these algorithms are trained to mirror *a population of users* and therefore there is inherent tension between their individual and collective performance. In particular, in recent years X has increasingly experienced a shift toward the right of the political spectrum: in an analysis from October 2024 (concurrent with our study), a Washington Post analysis found that following Elon Musk's takeover of X, there have emerged partisan differences in user base and engagement [8]. During this period, virtually all of the viral posts from congressional accounts came from Republican accounts, and Republican accounts accumulated more follower counts and views than Democrat accounts. We note that according to the 2024 National Public Opinion Reference Survey (conducted by Pew Research fielded between February 1 and June 10, 2024), X was still majority Democrats as of our study (48.4% Republican and 51.59% Democrat, [8])).

We focus our inquiry on the basic values associated with algorithmic amplification, but future work can use this same method to similarly evaluate these algorithms for political content, misinformation, or other social constructs. We note that our method measures amplification for in-network content, where we observe the entire inventory. It does not capture the amplification of out-of-network content where we cannot observe the inventory of content that could have been recommended. However, since out-of-network content, due to its sheer volume, is often selected for engagement across a broad audience, we hypothesize that these in-network effects would extend (and perhaps be even stronger) for out-of- network content (2). Furthermore, we note that this field study was conducted in September and October 2024 before X rolled out their Grok-based ranking algorithm (14) and before the 2024 US presidential election. Future work should extend this analysis to longitudinal snapshots across architectural changes to the platform to understand their relative effects on downstream user alignment. We also note that our framework for measuring values and alignment operates at the post-level, and may not reflect distributional feed-level value outcomes. Future work might explore more long-term or integrative methods for evaluating value alignment with feed algorithms.

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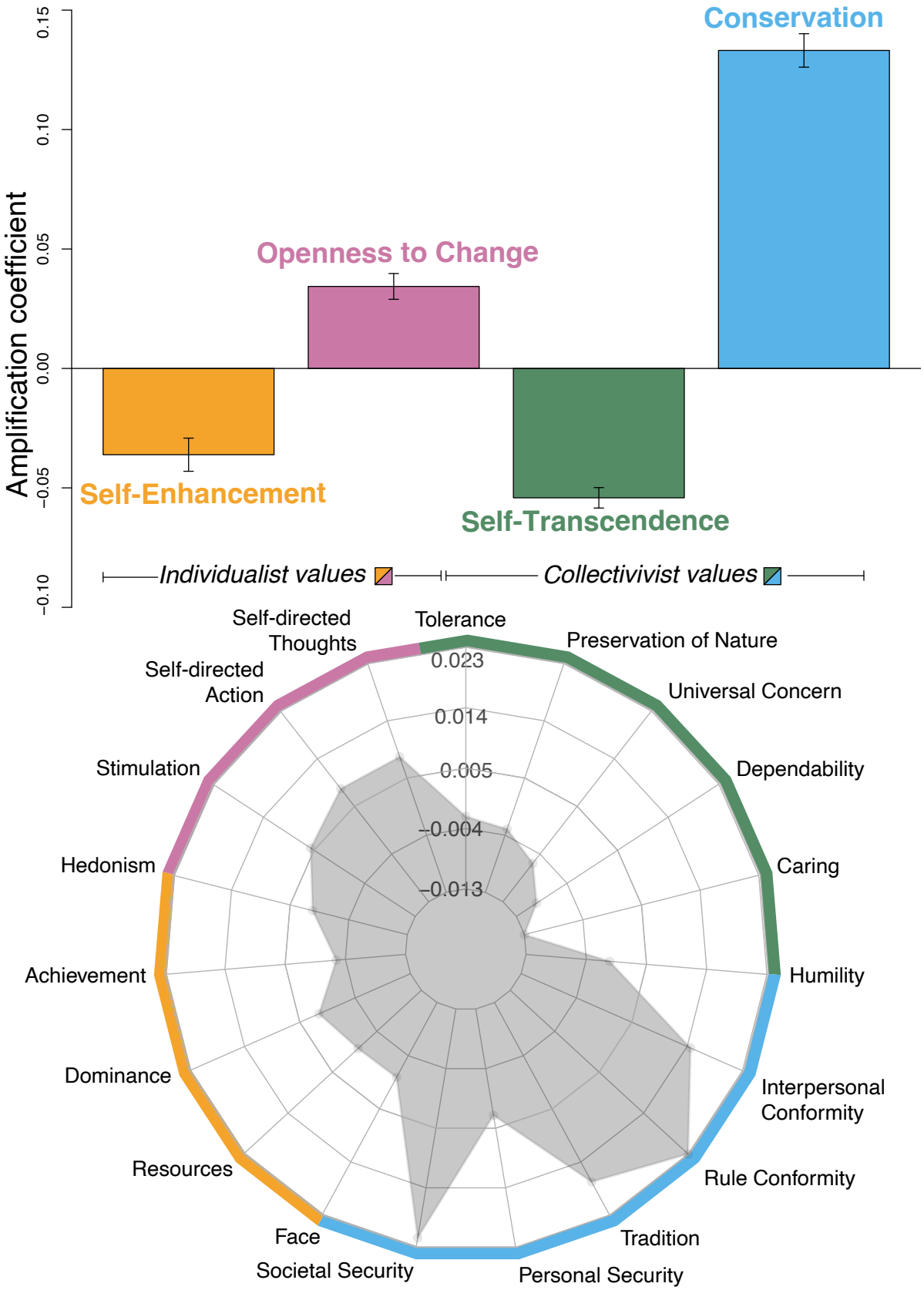

Figure 1: Top: Amplification results for the four middle-level value clusters of *Self-transcendence*, *Openness to change*, *Conservation*, *Self-enhancement*. Bottom: a radar plot of amplification results for low-level values using coefficients computed from the pre-registered PCA analysis (see Materials and Methods).

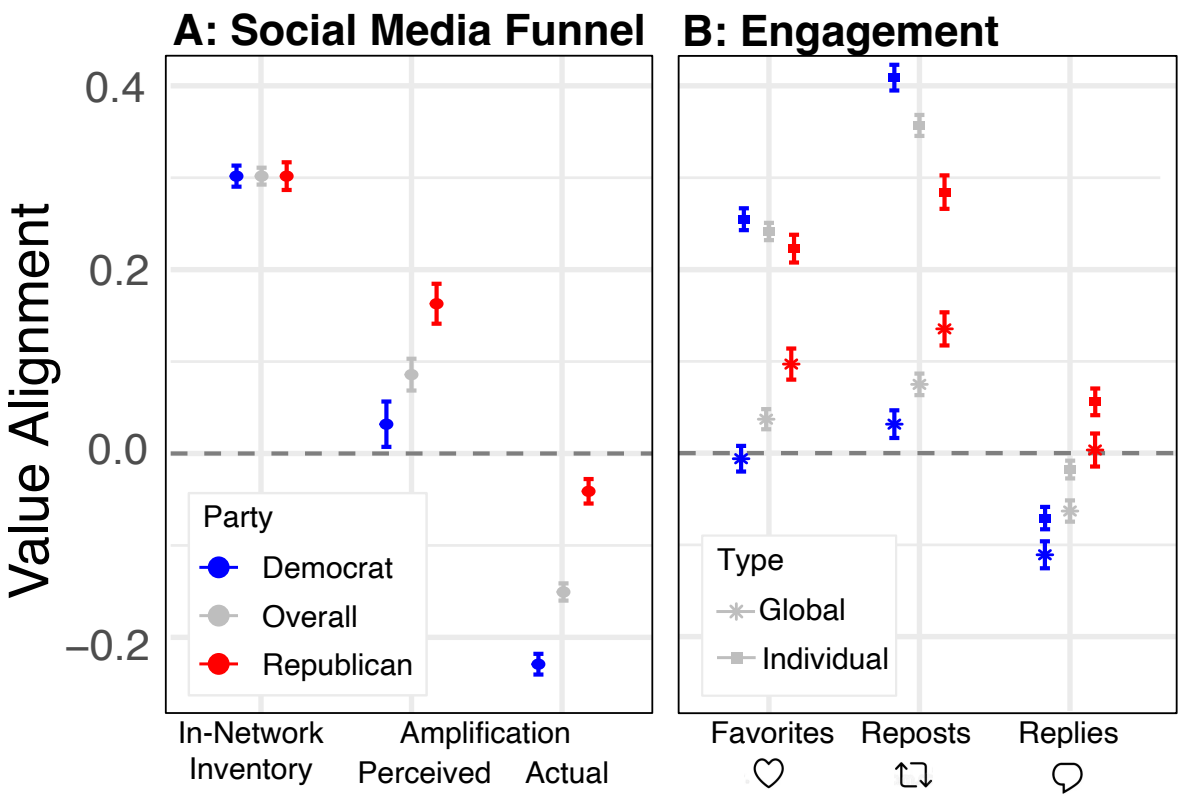

Figure 2: Value alignment (Spearman rank correlation with personal values) for Democrats (Blue), Republicans (Red) and Overall (Grey). A: Value alignment for in-network inventory, amplification (see Fig. 1) and perceived amplification. B: Value alignment for values associated with participant's own engagement patterns up to 24 hours after the study (square) and global engagement across the entire platform (overall number of replies, retweets and favorites that posts accrued, star).